

Homework 1

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B1.

- (i) **Setup.** We use the vector space axioms as labeled in the text: E is an abelian group under $+$, meaning $+$ satisfies (G1) associativity, (G2) existence of an identity 0 , (G3) existence of inverses, and (Ab) commutativity. Scalar multiplication satisfies (V1) $\alpha \cdot (u + v) = \alpha \cdot u + \alpha \cdot v$, (V2) $(\alpha + \beta) \cdot v = \alpha \cdot v + \beta \cdot v$, (V3) $(\alpha\beta) \cdot v = \alpha \cdot (\beta \cdot v)$, and (V4) $1 \cdot v = v$.

Claim 1: $\alpha \cdot 0 = 0$.

$$\begin{array}{ll}
 0 = 0 + 0 & \text{(G2, taking } v = 0\text{)} \\
 \alpha \cdot 0 = \alpha \cdot (0 + 0) & \text{(substituting the line above)} \\
 \alpha \cdot 0 = \alpha \cdot 0 + \alpha \cdot 0 & \text{(V1)} \\
 \alpha \cdot 0 + (-(\alpha \cdot 0)) = (\alpha \cdot 0 + \alpha \cdot 0) + (-(\alpha \cdot 0)) & \text{(adding } -(\alpha \cdot 0) \text{ to both sides)} \\
 0 = (\alpha \cdot 0 + \alpha \cdot 0) + (-(\alpha \cdot 0)) & \text{(G3, on the left side)} \\
 0 = \alpha \cdot 0 + (\alpha \cdot 0 + (-(\alpha \cdot 0))) & \text{(G1, regrouping the right side)} \\
 0 = \alpha \cdot 0 + 0 & \text{(G3, on the inner term)} \\
 0 = \alpha \cdot 0 & \text{(G2)}
 \end{array}$$

Hence $\alpha \cdot 0 = 0$.

Claim 2: $0 \cdot v = 0$ (the 0 on the left is the zero scalar in K).

$$\begin{array}{ll}
 0 = 0 + 0 & \text{(zero scalar in } K: 0 + 0 = 0\text{)} \\
 0 \cdot v = (0 + 0) \cdot v & \text{(substituting the line above)} \\
 0 \cdot v = 0 \cdot v + 0 \cdot v & \text{(V2)} \\
 0 \cdot v + (-(0 \cdot v)) = (0 \cdot v + 0 \cdot v) + (-(0 \cdot v)) & \text{(adding } -(0 \cdot v) \text{ to both sides)} \\
 0 = (0 \cdot v + 0 \cdot v) + (-(0 \cdot v)) & \text{(G3, on the left side)} \\
 0 = 0 \cdot v + (0 \cdot v + (-(0 \cdot v))) & \text{(G1, regrouping the right side)} \\
 0 = 0 \cdot v + 0 & \text{(G3, on the inner term)} \\
 0 = 0 \cdot v & \text{(G2)}
 \end{array}$$

Hence $0 \cdot v = 0$.

Claim 3: $\alpha \cdot (-v) = -(\alpha \cdot v)$.

$$\begin{aligned}
v + (-v) &= 0 && \text{(G3)} \\
\alpha \cdot (v + (-v)) &= \alpha \cdot 0 && \text{(applying } \alpha \cdot (\cdot) \text{ to both sides)} \\
\alpha \cdot v + \alpha \cdot (-v) &= \alpha \cdot 0 && \text{(V1, on the left side)} \\
\alpha \cdot v + \alpha \cdot (-v) &= 0 && \text{(Claim 1)} \\
-(\alpha \cdot v) + (\alpha \cdot v + \alpha \cdot (-v)) &= -(\alpha \cdot v) + 0 && \text{(adding } -(\alpha \cdot v) \text{ to both sides)} \\
(-(\alpha \cdot v) + \alpha \cdot v) + \alpha \cdot (-v) &= -(\alpha \cdot v) + 0 && \text{(G1, regrouping the left side)} \\
0 + \alpha \cdot (-v) &= -(\alpha \cdot v) + 0 && \text{(G3, on the left side)} \\
0 + \alpha \cdot (-v) &= -(\alpha \cdot v) && \text{(G2, on the right side)} \\
\alpha \cdot (-v) &= -(\alpha \cdot v) && \text{(G2, on the left side)}
\end{aligned}$$

Hence $\alpha \cdot (-v) = -(\alpha \cdot v)$.

Claim 4: $(-\alpha) \cdot v = -(\alpha \cdot v)$.

$$\begin{aligned}
\alpha + (-\alpha) &= 0 && \text{(additive inverse of the scalar } \alpha \text{ in } K) \\
(\alpha + (-\alpha)) \cdot v &= 0 \cdot v && \text{(applying } (\cdot) \cdot v \text{ to both sides)} \\
\alpha \cdot v + (-\alpha) \cdot v &= 0 \cdot v && \text{(V2, on the left side)} \\
\alpha \cdot v + (-\alpha) \cdot v &= 0 && \text{(Claim 2)} \\
-(\alpha \cdot v) + (\alpha \cdot v + (-\alpha) \cdot v) &= -(\alpha \cdot v) + 0 && \text{(adding } -(\alpha \cdot v) \text{ to both sides)} \\
(-(\alpha \cdot v) + \alpha \cdot v) + (-\alpha) \cdot v &= -(\alpha \cdot v) + 0 && \text{(G1, regrouping the left side)} \\
0 + (-\alpha) \cdot v &= -(\alpha \cdot v) + 0 && \text{(G3, on the left side)} \\
0 + (-\alpha) \cdot v &= -(\alpha \cdot v) && \text{(G2, on the right side)} \\
(-\alpha) \cdot v &= -(\alpha \cdot v) && \text{(G2, on the left side)}
\end{aligned}$$

Hence $(-\alpha) \cdot v = -(\alpha \cdot v)$.

- (ii) **Lemma (V1 extends to finite sums).** Let $\cdot : \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ be an operation satisfying (V1): $\alpha \cdot (u+v) = \alpha \cdot u + \alpha \cdot v$ for all $\alpha \in \mathbb{R}$ and $u, v \in \mathbb{R}^n$. Then for any $m \geq 1$, any vectors $u_1, \dots, u_m \in \mathbb{R}^n$, and any $\alpha \in \mathbb{R}$,

$$\alpha \cdot (u_1 + u_2 + \dots + u_m) = \alpha \cdot u_1 + \alpha \cdot u_2 + \dots + \alpha \cdot u_m.$$

Proof. We proceed by induction on m .

Base case ($m = 1$): The statement reads $\alpha \cdot u_1 = \alpha \cdot u_1$, which holds trivially.

Induction step: Let $k \geq 1$ be arbitrary and assume (IH) that

$$\alpha \cdot (u_1 + \dots + u_k) = \alpha \cdot u_1 + \dots + \alpha \cdot u_k$$

holds for any k vectors $u_1, \dots, u_k \in \mathbb{R}^n$. We show the statement holds for $k + 1$ vectors $u_1, \dots, u_k, u_{k+1}$.

$$\begin{aligned}
u_1 + \cdots + u_k + u_{k+1} &= (u_1 + \cdots + u_k) + u_{k+1} && \text{(grouping the first } k \text{ terms)} \\
\alpha \cdot (u_1 + \cdots + u_k + u_{k+1}) &= \alpha \cdot ((u_1 + \cdots + u_k) + u_{k+1}) && \text{(substituting the line above)} \\
&= \alpha \cdot (u_1 + \cdots + u_k) + \alpha \cdot u_{k+1} && \text{(V1, applied to the two terms } (u_1 + \cdots + u_k) \text{ and } u_{k+1}) \\
&= (\alpha \cdot u_1 + \cdots + \alpha \cdot u_k) + \alpha \cdot u_{k+1} && \text{(IH)}
\end{aligned}$$

Hence $\alpha \cdot (u_1 + \cdots + u_k + u_{k+1}) = \alpha \cdot u_1 + \cdots + \alpha \cdot u_k + \alpha \cdot u_{k+1}$, completing the induction. $\square$

Applying the Lemma. Recall that every $x = (x_1, \dots, x_n) \in \mathbb{R}^n$ can be written uniquely as

$$x = x_1 e_1 + x_2 e_2 + \cdots + x_n e_n,$$

where $e_i = (0, \dots, 0, 1, 0, \dots, 0)$ has a single 1 in position i . Applying the Lemma with $m = n$ and $u_i = x_i e_i$ for $i = 1, \dots, n$, we get, for any $\alpha \in \mathbb{R}$,

$$\alpha \cdot x = \alpha \cdot (x_1 e_1 + x_2 e_2 + \cdots + x_n e_n) = \alpha \cdot (x_1 e_1) + \alpha \cdot (x_2 e_2) + \cdots + \alpha \cdot (x_n e_n),$$

as required.

Conclusion. The right-hand side above depends only on the values $\alpha \cdot (x_1 e_1), \dots, \alpha \cdot (x_n e_n)$. Each term $\alpha \cdot (x_i e_i)$ is the result of applying $\cdot$ to the scalar α and a vector $x_i e_i$ lying in the one-dimensional subspace spanned by e_i (namely $\{t e_i \mid t \in \mathbb{R}\}$). Thus, once the action of $\cdot$ on each subspace $\mathbb{R} \cdot e_i$ (i.e., the value of $\alpha \cdot (t e_i)$ for every $\alpha, t \in \mathbb{R}$) is known, the value of $\alpha \cdot x$ for *every* $x \in \mathbb{R}^n$ is completely determined via the identity above. In other words, $\cdot$ is completely determined by its action on the one-dimensional subspaces spanned by $e_1, \dots, e_n$.

(iii) **Construction.** Fix an index k with $1 \leq k \leq n$. Define $\cdot : \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ as follows: for $\alpha \in \mathbb{R}$ and $x = (x_1, \dots, x_n) \in \mathbb{R}^n$, using the decomposition $x = x_1 e_1 + \cdots + x_n e_n$ from part (ii), set

$$\alpha \cdot x := \alpha \cdot (x_1 e_1) + \cdots + \alpha \cdot (x_n e_n), \quad \text{where } \alpha \cdot (x_i e_i) := \begin{cases} 0 & \text{if } i = k \\ (\alpha x_i) e_i & \text{if } i \neq k. \end{cases}$$

In words: $\cdot$ acts as ordinary scalar multiplication on every coordinate except coordinate k , which it always sends to 0, regardless of α .

Verification of (V1): $\alpha \cdot (u + v) = \alpha \cdot u + \alpha \cdot v$.

Let $u = (u_1, \dots, u_n)$ and $v = (v_1, \dots, v_n)$, so $u + v = (u_1 + v_1, \dots, u_n + v_n)$.

$$\begin{aligned}
\alpha \cdot (u + v) &= \sum_{i \neq k} \alpha(u_i + v_i) e_i && \text{(definition of } \cdot) \\
&= \sum_{i \neq k} (\alpha u_i) e_i + \sum_{i \neq k} (\alpha v_i) e_i \\
&= \alpha \cdot u + \alpha \cdot v && \text{(definition of } \cdot)
\end{aligned}$$

Hence (V1) holds.

Verification of (V2): $(\alpha + \beta) \cdot v = \alpha \cdot v + \beta \cdot v$.

Let $v = (v_1, \dots, v_n)$.

$$\begin{aligned}
 (\alpha + \beta) \cdot v &= \sum_{i \neq k} (\alpha + \beta) v_i e_i && \text{(definition of } \cdot \text{)} \\
 &= \sum_{i \neq k} (\alpha v_i) e_i + \sum_{i \neq k} (\beta v_i) e_i \\
 &= \alpha \cdot v + \beta \cdot v && \text{(definition of } \cdot \text{)}
 \end{aligned}$$

Hence (V2) holds.

Verification of (V3): $(\alpha\beta) \cdot v = \alpha \cdot (\beta \cdot v)$.

Let $v = (v_1, \dots, v_n)$. We check both sides coordinate by coordinate.

Coordinate k : By definition, $(\alpha\beta) \cdot v$ has 0 in coordinate k . On the other side, $\beta \cdot v$ has 0 in coordinate k , and applying $\alpha \cdot (\cdot)$ leaves coordinate k equal to 0 as well. So both sides agree ($= 0$).

Coordinate $i \neq k$:

$$\begin{aligned}
 [(\alpha\beta) \cdot v]_i &= (\alpha\beta) v_i && \text{(definition of } \cdot \text{)} \\
 &= \alpha(\beta v_i) \\
 &= \alpha \cdot [\beta \cdot v]_i \\
 &= [\alpha \cdot (\beta \cdot v)]_i
 \end{aligned}$$

So both sides agree in every coordinate. Hence $(\alpha\beta) \cdot v = \alpha \cdot (\beta \cdot v)$, and (V3) holds.

Failure of (V4). Recall (V4) requires $1 \cdot v = v$ for all $v \in \mathbb{R}^n$. Take $v = e_k$, so $v_k = 1$ and $v_i = 0$ for $i \neq k$.

$$1 \cdot e_k = \sum_{i \neq k} (1 \cdot 0) e_i = 0.$$

But $e_k \neq 0$. Hence $1 \cdot e_k = 0 \neq e_k$, so (V4) fails for this operation.

Conclusion. The operation $\cdot$ defined above satisfies (V1), (V2), and (V3), but not (V4). This shows that (V4) is not a consequence of (V1)–(V3): it must be imposed as an independent axiom.

(iv) **int lemma:** we will prove this for integers r first.

$$\begin{aligned}
 r \cdot x &= (1 + \dots + 1) \cdot x && (r \text{ times}) \\
 &= 1 \cdot x + \dots + 1 \cdot x && \text{(repeated distributive property)} \\
 &= r(1 \cdot x)
 \end{aligned}$$

now we prove for rationals: $r \in \mathbb{Q}$ implies that $r = p/q$, for some $p, q \in \mathbb{Z}$ and $q \neq 0$. We consider the quantity $p \cdot x$.

$$\begin{aligned}
 (q \times \frac{p}{q}) \cdot x &= (q \times r) \cdot x && = q \cdot (r \cdot x) && \text{(V3)} \\
 p \cdot x &= && = q(r \cdot x) && \text{(int lemma)} \\
 p(1 \cdot x) &= && && \text{(int lemma)}
 \end{aligned}$$

So we have that $p(1 \cdot x) = q(r \cdot x) \implies \frac{p}{q}(1 \cdot x) = 1(r \cdot x) = 1 \cdot (r \cdot x) = (1 \times r) \cdot x = r \cdot x$ (V3). Since $\frac{p}{q} = r$, we have that $r(1 \cdot x) = r \cdot x$.

B2.

- (1) **Proposition.** Let $(u_1, \dots, u_m)$ be a linearly independent family in a vector space E , and let $v_i = \sum_{j=1}^m a_{ij}u_j$ for $i = 1, \dots, m$, where $A = (a_{ij})$ is upper triangular with $a_{ii} \neq 0$ for all i . Then $(v_1, \dots, v_m)$ is linearly independent.

Proof. By induction on m .

Base case ($m = 1$). Here $v_1 = a_{11}u_1$ with $a_{11} \neq 0$ and $u_1 \neq 0$ (as (u_1) is independent), so $v_1 \neq 0$. Hence (v_1) is independent.

Induction step. Assume the proposition holds for families of size k ; we prove it for $k + 1$. Suppose

$$c_1v_1 + c_2v_2 + \dots + c_{k+1}v_{k+1} = 0.$$

Since A is upper triangular, u_1 occurs only in v_1 (as $a_{i1} = 0$ for $i > 1$), so the coefficient of u_1 in the expansion of the left-hand side is c_1a_{11} . By independence of $u_1, \dots, u_{k+1}$, this coefficient must vanish, and since $a_{11} \neq 0$, we get $c_1 = 0$.

The equation reduces to $c_2v_2 + \dots + c_{k+1}v_{k+1} = 0$. Each v_i ($i \geq 2$) involves only $u_2, \dots, u_{k+1}$, which are independent (a subfamily of an independent family), and the submatrix $(a_{ij})_{2 \leq i, j \leq k+1}$ is upper triangular with nonzero diagonal. By the induction hypothesis, $v_2, \dots, v_{k+1}$ are independent, so $c_2 = \dots = c_{k+1} = 0$.

Thus all $c_i = 0$, proving $(v_1, \dots, v_{k+1})$ independent and completing the induction.

- (2) **Proposition.** Let $A = (a_{ij})$ be an upper-triangular $n \times n$ matrix with $a_{ii} \neq 0$ for all i . Then A is invertible, and A^{-1} is also upper triangular.

Proof. By induction on n , using block multiplication.

Base case ($n = 1$). $A = (a_{11})$ with $a_{11} \neq 0$, so $A^{-1} = (a_{11}^{-1})$, trivially upper triangular.

Induction step. Assume every $(n - 1) \times (n - 1)$ upper-triangular matrix with nonzero diagonal is invertible with upper-triangular inverse. Write A in block form

$$A = \begin{pmatrix} A' & b \\ 0 & a_{nn} \end{pmatrix},$$

where A' is the upper-triangular $(n - 1) \times (n - 1)$ leading block with nonzero diagonal, b is $(n - 1) \times 1$, and $a_{nn} \neq 0$. By the induction hypothesis, A' is invertible with upper-triangular inverse $B' = A'^{-1}$. Set $d = a_{nn}^{-1}$ and $c = -d B'b$, and define

$$B = \begin{pmatrix} B' & c \\ 0 & d \end{pmatrix},$$

which is upper triangular since B' is.

Compute

$$AB = \begin{pmatrix} A'B' & A'c + bd \\ 0 & a_{nn}d \end{pmatrix} = \begin{pmatrix} I_{n-1} & -d(A'B')b + bd \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} I_{n-1} & 0 \\ 0 & 1 \end{pmatrix} = I_n,$$

using $A'B' = I_{n-1}$. Similarly,

$$BA = \begin{pmatrix} B'A' & B'b + ca_{nn} \\ 0 & da_{nn} \end{pmatrix} = \begin{pmatrix} I_{n-1} & B'b - B'b \\ 0 & 1 \end{pmatrix} = I_n,$$

using $ca_{nn} = -d a_{nn} B'b = -B'b$. Hence $B = A^{-1}$, and B is upper triangular by construction. This completes the induction.

Proposition. Let $A = (a_{ij})$ be an upper-triangular $n \times n$ matrix. If A is invertible, then all diagonal entries of A are nonzero.

Proof. By induction on n , using block multiplication.

Base case ($n = 1$). $A = (a_{11})$ invertible means there exists b with $a_{11}b = 1$. If $a_{11} = 0$, then $a_{11}b = 0 \neq 1$, a contradiction. So $a_{11} \neq 0$.

Induction step. Assume the proposition holds for $(n - 1) \times (n - 1)$ upper-triangular invertible matrices. Let A be $n \times n$, upper triangular, invertible, with inverse B . Write both in block form

$$A = \begin{pmatrix} A' & b \\ 0 & a_{nn} \end{pmatrix}, \quad B = \begin{pmatrix} P & q \\ r & s \end{pmatrix},$$

where A' is the $(n - 1) \times (n - 1)$ upper-triangular leading block of A , and B is partitioned to match (no structure on B is assumed yet). Then

$$AB = \begin{pmatrix} A'P + br & A'q + bs \\ a_{nn}r & a_{nn}s \end{pmatrix} = \begin{pmatrix} I_{n-1} & 0 \\ 0 & 1 \end{pmatrix} = I_n.$$

Comparing bottom-row entries gives two equations:

$$a_{nn}r = 0, \quad a_{nn}s = 1.$$

If $a_{nn} = 0$, the second equation reads $0 = 1$, a contradiction; hence $a_{nn} \neq 0$. Substituting into the first equation, $a_{nn}r = 0$ then forces $r = 0$. Comparing the top-left entries of $AB = I_n$, and using $r = 0$, gives

$$A'P + br = A'P = I_{n-1}.$$

An entirely analogous computation for $BA = I_n$ (comparing top-left entries) gives $PA' = I_{n-1}$.

Thus $A'P = PA' = I_{n-1}$, so A' is invertible. Since A' is upper triangular and invertible, the induction hypothesis gives that all diagonal entries $a_{11}, \dots, a_{n-1, n-1}$ of A' are nonzero. Combined with $a_{nn} \neq 0$, every diagonal entry of A is nonzero. This completes the induction

- (3) **Proposition.** Let $(u_1, \dots, u_m)$, $(v_1, \dots, v_m)$, and $A = (a_{ij})$ be as in (1), with A upper triangular and all diagonal entries nonzero. Then $(u_1, \dots, u_m)$ is linearly independent if and only if $(v_1, \dots, v_m)$ is.

Proof.

($\Rightarrow$) This is exactly the statement of (1).

($\Leftarrow$) By (2), A is invertible with upper-triangular inverse A^{-1} . Also by (2), since A^{-1} is upper triangular and invertible (its inverse is A), its diagonal entries are all nonzero.

Writing $v_i = \sum_j a_{ij}u_j$ in matrix form as $V = AU$, invertibility of A gives $U = A^{-1}V$, i.e.

$$u_i = \sum_j (A^{-1})_{ij} v_j, \quad i = 1, \dots, m.$$

This is precisely the setup of (1), with the roles of (u_i) and (v_i) exchanged and A replaced by A^{-1} (upper triangular, nonzero diagonal). Assuming $(v_1, \dots, v_m)$ is linearly independent, (1) applied in this form gives that $(u_1, \dots, u_m)$ is linearly independent.

Combining both directions proves the proposition.

B3.

(1) Expanding the matrix into a system of equations, we have that

$$\begin{aligned} b_1 &= x_1 + 2x_2 \\ b_2 &= x_2 + 2x_3 \\ &\vdots \\ b_{n-1} &= x_{n-1} + 2x_n \\ b_n &= x_n \end{aligned}$$

Isolating the x_k term, we can rearrange to get

$$\begin{aligned} x_1 &= b_1 - 2x_2 \\ x_2 &= b_2 - 2x_3 \\ &\vdots \\ x_{n-1} &= b_{n-1} - 2x_n \\ x_n &= b_n \end{aligned}$$

Finally, we claim that $x_k = \sum_{i=k}^n (-2)^{i-k} b_i$.

Proof: [reverse] Induction on k .

Base case $k = n$: $x_n = \sum_{i=n}^n (-2)^{i-n} b_i = b_n$.

IH: Assume for some $k \in [2..n]$ that $x_k = \sum_{i=k}^n (-2)^{i-k} b_i$.

IS: We wish to show $x_{k-1} = \sum_{i=k-1}^n (-2)^{i-(k-1)} b_i$.

$$\begin{aligned} x_{k-1} &= b_{k-1} - 2x_k && \text{(is defined bc } k \geq 2) \\ &= b_{k-1} - 2 \sum_{i=k}^n (-2)^{i-k} b_i && \text{(IH)} \\ &= b_{k-1} + \sum_{i=k}^n (-2)^{i-k+1} b_i \\ &= (-2)^{(k-1)-(k-1)} b_{k-1} + \sum_{i=k}^n (-2)^{i-(k-1)} b_i \\ &= \sum_{i=k-1}^n (-2)^{i-(k-1)} b_i \end{aligned}$$

As such we have proved the claim holds for $k \in [1..n]$.

So the solution is $x = (x_1, \dots, x_n)$ where $x_k = \sum_{i=k}^n (-2)^{i-k} b_i$

(2) Notably, since we've just computed x in terms of b , we can express $x = A^{-1}b$ where $A_{ij}^{-1} = \mathbb{1}\{j \geq i\}(-2)^{j-i}$, since $x_k = \sum_{i=k}^n (-2)^{i-k} b_i = \sum_{i=1}^n \mathbb{1}\{i \geq k\}(-2)^{i-k} b_i = \sum_i A_{ki}^{-1} b_i$. From a matrix perspective, this is saying that

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{pmatrix} = \begin{bmatrix} 1 & -2 & 4 & \cdots & (-2)^{n-1} \\ 0 & 1 & -2 & \cdots & (-2)^{n-2} \\ 0 & 0 & 1 & \cdots & (-2)^{n-3} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix} \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \\ b_n \end{pmatrix}$$

We claim that this matrix A^{-1} is the inverse of A .

Proof: $A^{-1}A = I$:

Let $C = A^{-1}A$. Looking at the diagonal, $C_{ii} = \sum_{k=1}^n A_{ik}^{-1}A_{ki} = \sum_{k \in \{i-1, i\}} \mathbb{1}\{k \geq i\}(-2)^{k-i}A_{ki} = 0 \cdot 2 + 1 \cdot 1 = 1$. Note that if $i = 1$, the $k = i - 1$ term does not exist, but that's okay since the indicator would've wiped it anyways.

For the non-diagonal term:

$$\begin{aligned} C_{ij} &= \sum_{k=1}^n A_{ik}^{-1}A_{kj} \\ &= A_{i(j-1)}^{-1}A_{(j-1)j} + A_{ij}^{-1}A_{jj} \\ &= 2A_{i(j-1)}^{-1} + A_{ij}^{-1} \\ &= 2 \cdot \mathbb{1}\{j-1 \geq i\}(-2)^{j-1-i} + \mathbb{1}\{j \geq i\}(-2)^{j-i} \\ &= \mathbb{1}\{j \geq i\}(2 \cdot (-2)^{j-1-i} + (-2)^{j-i}) && (\text{since } j \neq i, \text{ we have } j-1 \geq i \iff j \geq i) \\ &= \mathbb{1}\{j \geq i\}(-(-2)^{j-i} + (-2)^{j-i}) \\ &= 0 \end{aligned}$$

So $C = I$.

Proof: $AA^{-1} = I$:

Let $D = AA^{-1}$. For the diagonal terms, $D_{ii} = \sum_{k=1}^n A_{ik}A_{ki}^{-1} = \sum_{k=1}^n A_{ik}A_{ki}^{-1} = A_{ii}A_{ii}^{-1} + A_{i(i+1)}A_{(i+1)i}^{-1} = 1 \cdot 1 + 2 \cdot 0 = 1$.

Looking at the non-diagonal terms,

$$\begin{aligned} D_{ij} &= \sum_{k=1}^n A_{ik}A_{kj}^{-1} \\ &= A_{ii}A_{ij}^{-1} + A_{i(i+1)}A_{(i+1)j}^{-1} \\ &= A_{ij}^{-1} + 2A_{(i+1)j}^{-1} \\ &= \mathbb{1}\{j \geq i\}(-2)^{j-i} + 2 \cdot \mathbb{1}\{j \geq i+1\}(-2)^{j-i-1} \\ &= \mathbb{1}\{j \geq i\}((-2)^{j-i} + 2 \cdot (-2)^{j-i-1}) && (\text{since } j \neq i, \text{ we have } j \geq i+1 \iff j \geq i) \\ &= \mathbb{1}\{j \geq i\}((-2)^{j-i} - (-2)^{j-i}) \\ &= 0 \end{aligned}$$

So $D = I$.

So A^{-1} is the inverse of A .

We can note the largest magnitude coefficient of A^{-1} lies in the top right corner, specifically, $\max_{ij} |A_{ij}| = 2^{n-1}$. For $n \geq 300$, $2^{299} \approx 10^{90} > 10^{82}$. So the coefficients in this matrix exceed the size of the universe for $n \geq 300$.

- (3) We define $P(n)$ as $(A - I)^n$'s first n columns are zero columns. Proving $P(n)$ for $n \times n$ matrix A implies that $(A - I)^n$ is zero.

Induction on n .

Base Case $n = 1$. $(A - I)$ is a matrix with all zeros except for the second diagonal (one above the main diagonal) which is all 2s. The first column is all zeros.

Induction Hypothesis $n = k$: Assume $P(k)$ is true for some $k \geq 1$ and $k \leq n - 1$.

Induction step: Observe that $(A - I)^{k+1} = (A - I)^k(A - I)$. Let's consider $(A - I)_{ij}^{k+1}$ for any $j \leq k + 1$.

$$\begin{aligned}
 (A - I)_{ij}^{k+1} &= \sum_{z=1}^n (A - I)_{iz}^k (A - I)_{zj} \\
 &= (A - I)_{i(j-1)}^k (A - I)_{(j-1)j} && (A - I \text{ is only non-zero on the second diagonal}) \\
 &= 0 \cdot 2 && (j \leq k + 1 \implies j - 1 \leq k, \text{ by IH, } (A - I)_{iz}^k = 0 \text{ for any } z \leq j) \\
 &= 0
 \end{aligned}$$

So the first to $k + 1$ th column are all zeros.

Addendum: note that technically the indexing doesn't work if $j = 1$, but this just translates into summing across the 1st column of $A - I$, which is all zeros. So it's still 0.

B4.

(1) We can expand each of $B_i^2(t)$:

$$\begin{aligned}
 B_0^2(t) &= (1-t)^2 &= 1 - 2t + t^2 &= \begin{bmatrix} 1 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ t \\ t^2 \end{bmatrix} \\
 B_1^2(t) &= 2(1-t)t &= 2t - 2t^2 &= \begin{bmatrix} 0 & 2 & -2 \end{bmatrix} \begin{bmatrix} 1 \\ t \\ t^2 \end{bmatrix} \\
 B_2^2(t) &= t^2 &= t^2 &= \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ t \\ t^2 \end{bmatrix}
 \end{aligned}$$

This gives the desired representation of the Bernsteins in terms of the standard basis.

For this question (and the next question), observe that $B_k^m(t) = \binom{m}{k}(1-t)^{m-k}t^k$. As such, $\sum_{k=0}^m B_k^m(t) = \sum_{k=0}^m \binom{m}{k}(1-t)^{m-k}t^k = (t + (1-t))^m = 1^m = 1$, using the binomial expansion.

(2) We can expand each Bernstein again:

$$\begin{aligned}
 B_0^3(t) &= (1-t)^3 &= 1 - 3t + 3t^2 - t^3 &= \begin{bmatrix} 1 & -3 & 3 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ t \\ t^2 \\ t^3 \end{bmatrix} \\
 B_1^3(t) &= 3(1-t)^2t &= 3t - 6t^2 + 3t^3 &= \begin{bmatrix} 0 & 3 & -6 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ t \\ t^2 \\ t^3 \end{bmatrix} \\
 B_2^3(t) &= 3(1-t)t^2 &= 3t^2 - 3t^3 &= \begin{bmatrix} 0 & 0 & 3 & -3 \end{bmatrix} \begin{bmatrix} 1 \\ t \\ t^2 \\ t^3 \end{bmatrix} \\
 B_3^3(t) &= t^3 &= t^3 &= \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ t \\ t^2 \\ t^3 \end{bmatrix}
 \end{aligned}$$

This gives the desired representation of the Bernsteins in terms of the standard basis. We can use the proof above to note that the sum of the degree-3 bernsteins is also equal to 1.

- (3) Since the Bernstein polynomials form an upper triangular matrix transformation of the standard basis (which is linearly independent), (and have non-zero diagonal entries), (for degree 2 and 3), we can use problem B2's proof to state that the Bernstein polynomials are linearly independent.
- (4) By looking at a single $B_k^m(t)$ polynomial, we can decompose the polynomial into each coefficient of each power of t .

Let's arbitrarily say we are looking at power t^j for some j . We know $j \geq k$ because there's a power of t^k . So then there are $j - k$ remaining powers of t needed, which we can choose from the $m - k$ powers of $(1 - t)$. We have to choose $j - k$ from $m - k$ terms, of which there are $\binom{m-k}{j-k}$ ways to do so. There will also be a $(-1)^{j-k}$ term from the parity of $(1 - t)$.

So we now have that $B_k^m(t)_j$ (define this as the component of $B_k^m(t)$ that has power j of k), is equal to $\binom{m}{k}(-1)^{j-k}\binom{m-k}{j-k}$. We can simplify as:

$$\begin{aligned}
 B_k^m(t)_j &= \binom{m}{k}(-1)^{j-k}\binom{m-k}{j-k} \\
 &= (-1)^{j-k} \frac{m!}{k!(m-k)!} \frac{(m-k)!}{(j-k)!(m-k-(j-k))!} \\
 &= (-1)^{j-k} \frac{m!}{k!} \frac{1}{(j-k)!(m-j)!} \\
 &= (-1)^{j-k} \frac{m!}{(m-j)!j!} \frac{j!}{(j-k)!k!} \\
 &= (-1)^{j-k} \binom{m}{j} \binom{j}{k}
 \end{aligned}$$

Since $B_k^m(t) = \sum_j B_k^m(t)_j t^j$, we have that $B_k^m(t) = \sum_{j=k}^m (-1)^{j-k} \binom{m}{j} \binom{j}{k} t^j$. We note that the bounds of j go from k to m because:

- $j \geq k$ as mentioned previously because there is a power of t^k in the regular description of $B_k^m(t)$.
- $j \leq m$ because the total maximum power of t is only $(m-k)+k$, coming from the $(1-t)$ and t term respectively.

Since $B_k^m(t)$ can be expressed as the sum of t^j powers for $j \geq k$, the Bernstein polynomials can be rewritten as

$$\begin{bmatrix} B_0^m(t) \\ B_1^m(t) \\ \vdots \\ B_m^m(t) \end{bmatrix} = A \begin{bmatrix} 1 \\ t \\ \vdots \\ t^m \end{bmatrix}$$

which is an upper triangular matrix A with elements $A_{ij} = B_i^m(t)_j = (-1)^{j-i} \binom{m}{j} \binom{j}{i}$. Observe that if $i > j$, the last binomial term is zero, so it's upper triangular. Also note that $A_{ii} = (-1)^{i-i} \binom{m}{i} \binom{i}{i} = \binom{m}{i} > 0$. So the diagonal entries are non-zero.

Using problem B2, we know that the bernsteins form a linearly independent set.

(5) This is problem 2's proof, which I will copy down here:

$$\begin{aligned}
1 &= 1^m \\
&= (1 - t + t)^m \\
&= \sum_{k=0}^m \binom{m}{k} (1 - t)^{m-k} t^k && \text{(binomial theorem)} \\
&= \sum_{k=0}^m B_k^m(t)
\end{aligned}$$

Extra Credit: We can note that the matrix representing the transformation from the standard basis to the Bernstein basis is symmetric about the anti-diagonal. Specifically, this means that $A_{ij} = A_{(m+2-j)(m+2-i)}$. We will show this as follows:

$$\begin{aligned}
A_{(k+1)(j+1)} &= (-1)^{j-k} \binom{m}{j} \binom{j}{k} \\
&= (-1)^{k-j} \binom{m}{j} \binom{j}{k} \\
&= (-1)^{k-j+m-m} \frac{m!}{j!(m-j)!} \frac{j!}{k!(j-k)!} \\
&= (-1)^{m-j+k-m} \frac{m!}{(m-j)!} \frac{1}{k!(j-k)!} \\
&= (-1)^{m-j+(k-m)} \frac{m!}{k!} \frac{1}{(m-j)!(j-k)!} \\
&= (-1)^{m-j-(m-k)} \frac{m!}{k!(m-k)!} \frac{(m-k)!}{(m-j)!(j-k)!} \\
&= (-1)^{m-j-(m-k)} \binom{m}{m-k} \binom{m-k}{m-j} \\
&= A_{(m+1-j)(m+1-k)} \\
&= A_{(m+2-(j+1))(m+2-(k+1))}
\end{aligned}$$

So if $(i, j) = (k + 1, j + 1)$, we've shown that $A_{ij} = A_{(m+2-j)(m+2-i)}$, which shows that the matrix is symmetric about the anti-diagonal.

The sum of the coefficients of the $i + 1$ th row are the sum of the coefficients of t_j of B_i^m . This can be found as $B_i^m(1)$, because that reduces the standard polynomial basis to the vector of 1s, giving us the sum of the coefficients in the row. Since $B_i^m(1) = \binom{m}{i} (1 - t)^{m-i} t^i \big|_{t=1} = \binom{m}{i} 0^{m-i} = 0$, as long as $i \neq m$. Therefore, the sum of the coefficients on any row is 0 unless it's the last row.

On the last row, we have $B_m^m(1) = \binom{m}{m} (1 - 1)^0 1^m = 1$.

Since the matrix is symmetric about this anti-diagonal, the sum of the last column is equal to the sum of the first row, etc.

This implies that all column sums are equal to zero, unless it is the first column, as that corresponds to the last row.

In summary, the matrix is symmetric about the anti-diagonal and has row/column sums of zero, except for the first column and last row, which have sum 1.

(6) We see that

$$\begin{aligned}
\sum_{j=0}^{m-i} t^i B_j^{m-i}(t) &= t^i \sum_{j=0}^{m-i} B_j^{m-i}(t) \\
&= t^i \sum_{j=0}^{m-i} \binom{m-i}{j} (1-t)^{m-i-j} t^j \\
&= t^i ((1-t) + t)^{m-i} \quad (\text{binomial theorem}) \\
&= t^i
\end{aligned}$$

, as desired.

Second, we have that

$$\begin{aligned}
\binom{m}{i} \binom{m-i}{j} &= \frac{m!}{i!(m-i)!} \frac{(m-i)!}{j!(m-i-j)!} \\
&= \frac{m!}{i!} \frac{1}{j!(m-i-j)!} \\
&= \frac{m!}{i!(i+j)!} \frac{(i+j)!}{j!(m-i-j)!} \\
&= \frac{m!}{(m-i-j)!(i+j)!} \frac{(i+j)!}{j!i!} \\
&= \binom{m}{i+j} \binom{i+j}{i}
\end{aligned}$$

, as desired.

We can combine these to find that

$$\begin{aligned}
t^i &= \sum_{j=0}^{m-i} t^i B_j^{m-i}(t) \\
&= \sum_{j=0}^{m-i} t^i \binom{m-i}{j} (1-t)^{m-i-j} t^j \\
&= \sum_{j=0}^{m-i} t^i \frac{\binom{m}{i}}{\binom{m}{i}} \binom{m-i}{j} (1-t)^{m-i-j} t^j \\
&= \sum_{j=0}^{m-i} t^i \frac{\binom{m}{i+j} \binom{i+j}{i}}{\binom{m}{i}} (1-t)^{m-i-j} t^j \\
&= \sum_{j=0}^{m-i} \frac{\binom{i+j}{i}}{\binom{m}{i}} \binom{m}{i+j} (1-t)^{m-i-j} t^{i+j} \\
&= \sum_{j=0}^{m-i} \frac{\binom{i+j}{i}}{\binom{m}{i}} B_{i+j}^m(t)
\end{aligned}$$

, as desired.

Since the space of polynomials of degree $\leq m$ includes all t^i from $0 \leq i \leq m$, we have that t^i is made up of some linear combination of all the $B_k^m(t)$ s where $k = i + j$, notably, $j \leq m - i$ so $k \leq m$. Also, $k \geq 0$, because $i, j \geq 0$. So t^i is made up of some linear combination of $B_k^m(t)$ s from $0 \leq k \leq m$. Since any polynomial is made up of a linear combination of the standard polynomial basis, which itself is made up of a linear combination of the Bernstein Polynomials, and linear transformations of linear transforms are still linear, we have that any polynomial

can be made up of a linear combination of Bernstein Polynomials. This is exactly the definition of the Bernstein Polynomials forming a basis for the space of polynomials with degree $\leq m$.

To form the matrix A^{-1} that takes us from the Bernstein basis to the standard basis, we can just plug in values of i :

Degree ≤ 2 :

$$\begin{aligned} t^0 &= \sum_{j=0}^2 \frac{\binom{j}{0}}{\binom{2}{0}} B_j^2(t) &= B_0^2(t) + B_1^2(t) + B_2^2(t) \\ t^1 &= \sum_{j=0}^1 \frac{\binom{1+j}{1}}{\binom{2}{1}} B_{1+j}^2(t) &= \frac{1}{2} B_1^2(t) + B_2^2(t) \\ t^2 &= \sum_{j=0}^0 \frac{\binom{2+j}{2}}{\binom{2}{2}} B_{2+j}^2(t) &= B_2^2(t) \end{aligned}$$

, which tells us that $A_2^{-1} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1/2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$.

Degree ≤ 3 :

$$\begin{aligned} t^0 &= \sum_{j=0}^3 \frac{\binom{j}{0}}{\binom{3}{0}} B_j^3(t) &= B_0^3(t) + B_1^3(t) + B_2^3(t) + B_3^3(t) \\ t^1 &= \sum_{j=0}^2 \frac{\binom{1+j}{1}}{\binom{3}{1}} B_{1+j}^3(t) &= \frac{1}{3} B_1^3(t) + \frac{2}{3} B_2^3(t) + B_3^3(t) \\ t^2 &= \sum_{j=0}^1 \frac{\binom{2+j}{2}}{\binom{3}{2}} B_{2+j}^3(t) &= \frac{1}{3} B_2^3(t) + B_3^3(t) \\ t^3 &= \sum_{j=0}^0 \frac{\binom{3+j}{3}}{\binom{3}{3}} B_{3+j}^3(t) &= B_3^3(t) \end{aligned}$$

, which tells us that $A_3^{-1} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1/3 & 2/3 & 1 \\ 0 & 0 & 1/3 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$.

- (7) We note that $C(t) = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = \begin{pmatrix} \sum_{i=0}^{m_1} \alpha_i t^{m_1-i} \\ \sum_{i=0}^{m_2} \beta_i t^{m_2-i} \end{pmatrix}$ is a linear combination of t^k terms for $k \in [0..m]$ (where if $m_1, m_2 < m$, the t^k terms for $k > m_1$ or $k > m_2$ have coefficient 0. Since we have a linear transformation A^{-1} for polynomials of degree at most m to go from the standard basis to the Bernstein basis, then we also have that each of $x(t)$ and $y(t)$ can be represented as a linear combination of Bernstein polynomials with coefficients $b_0, \dots, b_m$.

More formally, $C(t) = \begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = \begin{pmatrix} \alpha^\top t \\ \beta^\top t \end{pmatrix}$ where $\alpha = [0 \ \dots \ 0 \ \alpha_{m_1} \ \dots \ \alpha_0]^\top$ of dimension $m+1$, β defined similarly, and $t = [1 \ t \ \dots \ t^m]^\top$. Since $t = A^{-1} B^m(t)$, where $B^m(t) = [B_0^m(t) \ B_1^m(t) \ \dots \ B_m^m(t)]^\top$, we have that $x(t) = \alpha^\top A^{-1} B^m(t)$ and $y(t) = \beta^\top A^{-1} B^m(t)$. This implies that $\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = \begin{pmatrix} \alpha^\top \\ \beta^\top \end{pmatrix} t = \begin{pmatrix} \alpha^\top \\ \beta^\top \end{pmatrix} A^{-1} B^m(t) =$

$PB^m(t)$ for $P = \begin{bmatrix} b_0 & b_1 & \cdots & b_m \end{bmatrix}$ and $P = \begin{pmatrix} \alpha^\top \\ \beta^\top \end{pmatrix} A^{-1}$.

We can then observe that $C(0) = \begin{pmatrix} x(0) \\ y(0) \end{pmatrix} = \sum_{i=0}^m b_i B_i^m(0) = \sum_{i=0}^m b_i \binom{m}{i} 1^{m-i} 0^i = b_0$. We can also observe that

$$C(1) = \begin{pmatrix} x(1) \\ y(1) \end{pmatrix} = \sum_{i=0}^m b_i B_i^m(1) = \sum_{i=0}^m b_i \binom{m}{i} 0^{m-i} 1^i = b_m.$$

Generally, the other points are not on the curve. To see this, say we have b_i (for $1 \leq i \leq m-1$) on the curve. Then we have that $\sum_{k=0}^m b_k B_k^m(t^*) = C(t^*) = b_i = b_i \sum_{k=0}^m B_k^m(t^*)$ for some t^* . But this means that $\sum_{k \neq i} b_k B_k^m(t^*) = \sum_{k \neq i} b_i B_k^m(t^*)$, which tells us that

$$\frac{\sum_{k \neq i} b_k B_k^m(t^*)}{\sum_{k \neq i} B_k^m(t^*)} = b_i$$

, in other words, b_i is a weighted average of the other points. This is not generally true, as many control points are often outside the convex hull of the other points in order for the curve to move throughout more of space.

B5.

(a)

$$\begin{aligned}
 Ax &= 0 && \text{(premise)} \\
 A^{-1}Ax &= A^{-1}0 && (A \text{ has an inverse}) \\
 I_n x &= 0 && \text{(inverse; multiplication by 0)} \\
 x &= 0
 \end{aligned}$$

, so $Ax = 0 \implies x = 0$ if A is invertible.

(b) For setup, note that $A(I_n - BA) = A - ABA = (I_m - AB)A$ and that $B(I_m - AB) = B - BAB = (I_n - BA)B$.

Note the contrapositive of the result from part a, which is “Let $A \in \mathbb{R}^{n \times n}$ be a square matrix. If there exists $x \in \mathbb{R}^n$ such that $Ax = 0$ and $x \neq 0$, then A is not invertible.” Call this property P for ease of use.

Also note the contrapositive of the converse of part a, which is “If $A \in \mathbb{R}^{n \times n}$ is not invertible, then there exists some $x \in \mathbb{R}^n$ such that $Ax = 0$ and $x \neq 0$.” Call this property P’ for ease of use.

Forwards: $I_m - AB$ invertible $\implies I_n - BA$ invertible:

We prove the contrapositive. Assume $I_n - BA$ is not invertible. By property P’, we have that $(I_n - BA)v = 0$ for some $v \neq 0$.

$$\begin{aligned}
 (I_n - BA)v &= 0 \\
 \implies A(I_n - BA)v &= 0 \\
 \implies (I_m - AB)Av &= 0
 \end{aligned}$$

Define $u = Av$. If $u \neq 0$, then we could use property P to say that $I_m - AB$ is not invertible. It suffices to show that $u = Av \neq 0$:

Assume for contradiction that $Av = 0$. Then $(I_n - BA)v = I_nv - BAv = v - B0 = v$, but we previously said that $(I_n - BA)v = 0$, so this implies that $v = 0$, a contradiction.

Backwards: $I_n - BA$ invertible $\implies I_m - AB$ invertible:

We prove the contrapositive. Assume $I_m - AB$ is not invertible. By property P’, we have that $(I_m - AB)v = 0$ for some $v \neq 0$.

$$\begin{aligned}
 (I_m - AB)v &= 0 \\
 \implies B(I_m - AB)v &= 0 \\
 \implies (I_n - BA)Bv &= 0
 \end{aligned}$$

Define $u = Bv$. If $u \neq 0$, then we could use property P to say that $I_n - BA$ is not invertible. It suffices to show that $u = Bv \neq 0$:

Assume for contradiction that $Bv = 0$. Then $(I_m - AB)v = I_mv - ABv = v - A0 = v$, but we previously said that $(I_m - AB)v = 0$, so this implies that $v = 0$, a contradiction.

Both directions have thus been proven.

B6.

- (1) **Proposition.** Let $n \geq 3$ and let $B = (b_{ij})$ be the $n \times n$ matrix of Problem B6, with columns $b_1, \dots, b_n \in \mathbb{R}^n$, and let $e_1, \dots, e_n$ be the canonical basis vectors of $\mathbb{R}^n$. Then

$$\begin{aligned} b_1 - b_j &= 2(e_1 + e_j), & j &= 2, \dots, n, \\ (n-3)b_1 - (b_2 + \dots + b_n) &= 2(n-2)e_1. \end{aligned}$$

Proof.

Column 1 of B has every entry equal to 1, so

$$b_1 = (1, 1, \dots, 1) = e_1 + e_2 + \dots + e_n. \quad (*)$$

Fix j with $2 \leq j \leq n$. Column j of B has entry -1 in positions 1 and j , and entry 1 elsewhere. Comparing with b_1 entrywise, $b_1 - b_j$ has entry 2 in positions 1 and j and entry 0 elsewhere, so

$$b_1 - b_j = 2(e_1 + e_j), \quad j = 2, \dots, n.$$

Add the $n-1$ identities $b_1 - b_j = 2(e_1 + e_j)$, for $j = 2, \dots, n$:

$$(b_1 - b_2) + (b_1 - b_3) + \dots + (b_1 - b_n) = 2(e_1 + e_2) + 2(e_1 + e_3) + \dots + 2(e_1 + e_n).$$

The left side has $n-1$ copies of b_1 , minus $b_2 + b_3 + \dots + b_n$:

$$(n-1)b_1 - (b_2 + b_3 + \dots + b_n).$$

The right side has $n-1$ copies of $2e_1$, plus $2e_2 + 2e_3 + \dots + 2e_n$:

$$2(n-1)e_1 + 2(e_2 + e_3 + \dots + e_n).$$

By (*), $e_2 + e_3 + \dots + e_n = b_1 - e_1$. Substituting into the right-hand side,

$$2(n-1)e_1 + 2(b_1 - e_1) = 2(n-2)e_1 + 2b_1.$$

Equating the two sides,

$$(n-1)b_1 - (b_2 + \dots + b_n) = 2(n-2)e_1 + 2b_1.$$

Subtracting $2b_1$ from both sides,

$$(n-3)b_1 - (b_2 + \dots + b_n) = 2(n-2)e_1.$$

Thus, we have proved the proposition

- (2) **Proposition.** Let $n \geq 3$ and let B be as in (1), with columns $b_1, \dots, b_n$. Then B is invertible, and its inverse $A = (a_{ij})$ is given by

$$\begin{aligned} a_{11} &= \frac{n-3}{2(n-2)}, & a_{i1} &= -\frac{1}{2(n-2)} \quad (2 \leq i \leq n), \\ a_{ii} &= -\frac{n-3}{2(n-2)} \quad (2 \leq i \leq n), & a_{ji} &= \frac{1}{2(n-2)} \quad (2 \leq i \leq n, j \neq i). \end{aligned}$$

Proof.

Strategy. For a square matrix, $BA = I_n$ suffices to conclude $A = B^{-1}$. Writing $BA = I_n$ column by column, $B \cdot (\text{column } i \text{ of } A) = e_i$ for each $i = 1, \dots, n$. Since $Bx = x_1b_1 + \dots + x_nb_n$ for any vector x , this says exactly: the entries of column i of A are the coefficients expressing e_i as a linear combination of $b_1, \dots, b_n$. It therefore suffices to find, for each i , such an expression.

Solving for e_1 . By the sum identity of (1),

$$(n-3)b_1 - (b_2 + \dots + b_n) = 2(n-2)e_1.$$

Dividing by $2(n-2)$,

$$e_1 = \frac{n-3}{2(n-2)}b_1 - \frac{1}{2(n-2)}(b_2 + \dots + b_n). \quad (\text{I})$$

Solving for e_j , $2 \leq j \leq n$. By the pairwise identity of (1), $b_1 - b_j = 2(e_1 + e_j)$, so

$$e_j = \frac{1}{2}b_1 - \frac{1}{2}b_j - e_1. \quad (\text{II})$$

Substitute (I) into (II):

$$e_j = \frac{1}{2}b_1 - \frac{1}{2}b_j - \frac{n-3}{2(n-2)}b_1 + \frac{1}{2(n-2)}(b_2 + \dots + b_n).$$

Collect the coefficient of b_1 :

$$\frac{1}{2} - \frac{n-3}{2(n-2)} = \frac{(n-2) - (n-3)}{2(n-2)} = \frac{1}{2(n-2)}.$$

Split $b_2 + \dots + b_n = b_j + \sum_{k=2, k \neq j}^n b_k$ and collect the coefficient of b_j :

$$-\frac{1}{2} + \frac{1}{2(n-2)} = \frac{-(n-2) + 1}{2(n-2)} = -\frac{n-3}{2(n-2)}.$$

Every remaining b_k ($k \geq 2, k \neq j$) keeps coefficient $\frac{1}{2(n-2)}$. Hence

$$e_j = \frac{1}{2(n-2)}b_1 - \frac{n-3}{2(n-2)}b_j + \frac{1}{2(n-2)} \sum_{k=2, k \neq j}^n b_k, \quad 2 \leq j \leq n. \quad (\text{III})$$

Assembling A . By the strategy above, (I) gives column 1 of A and (III) gives column j of A for $2 \leq j \leq n$:

$$\begin{aligned} \text{col}_1(A) : \quad a_{11} &= \frac{n-3}{2(n-2)}, \quad a_{i1} = -\frac{1}{2(n-2)} \quad (2 \leq i \leq n), \\ \text{col}_j(A), \quad 2 \leq j \leq n : \quad a_{1j} &= \frac{1}{2(n-2)}, \quad a_{jj} = -\frac{n-3}{2(n-2)}, \quad a_{kj} = \frac{1}{2(n-2)} \quad (k \geq 2, k \neq j). \end{aligned}$$

This is exactly the claimed formula for A (writing a_{ji} for $j \neq i$ in place of a_{1j} and a_{kj} above). Thus $BA = I_n$, and since B is square, B is invertible with $B^{-1} = A$. This proves the proposition.

- (3) **Proposition.** Let $n \geq 3$ and let $B = (B_{ki})$ be as in Problem B6, with columns $b_1, \dots, b_n$. For $i = 1, \dots, n$, let D_i be the $n \times n$ diagonal matrix with $(D_i)_{kk} = B_{ki}$ for $k = 1, \dots, n$ (and all off-diagonal entries 0). Then $(D_1, \dots, D_n)$ is a basis of the vector space $\mathcal{D}_n$ of $n \times n$ diagonal matrices.

Proof.

Linear independence. Suppose $c_1D_1 + \dots + c_nD_n = 0$ (the zero matrix), for scalars $c_1, \dots, c_n$. Since each D_i is diagonal, so is $c_1D_1 + \dots + c_nD_n$, and its (k, k) entry is

$$c_1B_{k1} + c_2B_{k2} + \dots + c_nB_{kn} = (Bc)_k, \quad c = (c_1, \dots, c_n).$$

Thus $c_1D_1 + \cdots + c_nD_n = 0$ if and only if $(Bc)_k = 0$ for every $k = 1, \dots, n$, i.e. $Bc = 0$. Since B is invertible, $Bc = 0$ forces $c = 0$ (by (a) of Problem B5, applied to B). Hence $c_1 = \cdots = c_n = 0$, and $(D_1, \dots, D_n)$ is linearly independent.

Spanning. Let $D \in \mathcal{D}_n$ be an arbitrary diagonal matrix, with $(D)_{kk} = d_k$ for $k = 1, \dots, n$, and set $d = (d_1, \dots, d_n)$. As in the independence argument, for any scalars $c_1, \dots, c_n$ the (k, k) entry of $c_1D_1 + \cdots + c_nD_n$ equals $(Bc)_k$; hence

$$c_1D_1 + \cdots + c_nD_n = D \iff Bc = d.$$

Since B is invertible by (2), $Bc = d$ has the solution $c = B^{-1}d$. With this choice, $(Bc)_k = d_k$ for every k , so $c_1D_1 + \cdots + c_nD_n = D$. Hence $(D_1, \dots, D_n)$ spans $\mathcal{D}_n$.

Being linearly independent and spanning, $(D_1, \dots, D_n)$ is a basis of $\mathcal{D}_n$.

B7.

- (1) **Proposition.** H , with matrix multiplication, is a group. The inverse of $M(a, b, c) := \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \in H$ is

$$M(a, b, c)^{-1} = M(-a, ac - b, -c).$$

H is not abelian.

Proof.

Multiplication law. For $M(a, b, c), M(a', b', c') \in H$,

$$M(a, b, c) \cdot M(a', b', c') = M(a + a', b + b' + ac', c + c'), \quad (*)$$

by direct computation of the matrix product (e.g. the $(1, 3)$ entry is $1 \cdot b' + a \cdot c' + b \cdot 1 = b' + ac' + b$; the other entries are checked the same way).

Closure. By $(*)$, the product of two elements of H is again of the form $M(\cdot, \cdot, \cdot)$, hence lies in H .

Associativity. Matrix multiplication is associative on all 3×3 matrices; since H is closed under multiplication, associativity holds on H .

Identity. I_3 is the identity element for multiplication on all 3×3 matrices, and $I_3 = M(0, 0, 0) \in H$ (take $a = b = c = 0$). Hence $I_3 \in H$ is the identity of H .

Inverses. Given $M(a, b, c) \in H$, seek $M(a', b', c')$ with $M(a, b, c) \cdot M(a', b', c') = M(0, 0, 0)$. By $(*)$, this requires $a + a' = 0$, $c + c' = 0$, and $b + b' + ac' = 0$, i.e.

$$a' = -a, \quad c' = -c, \quad b' = ac - b.$$

So $M(a, b, c)^{-1} := M(-a, ac - b, -c) \in H$. Checking the other order, by $(*)$ with $(a, b, c) \rightarrow (-a, ac - b, -c)$ and $(a', b', c') \rightarrow (a, b, c)$:

$$M(-a, ac - b, -c) \cdot M(a, b, c) = M(-a + a, (ac - b) + b + (-a)c, -c + c) = M(0, 0, 0).$$

So $M(a, b, c)^{-1}$ is a two-sided inverse, and it lies in H .

Since H is closed under an associative operation, has an identity, and every element has an inverse in H , H is a group.

Not abelian. Take $M(1, 0, 0)$ and $M(0, 0, 1)$. By $(*)$,

$$M(1, 0, 0) \cdot M(0, 0, 1) = M(1, 0 + 0 + 1 \cdot 1, 1) = M(1, 1, 1),$$

$$M(0, 0, 1) \cdot M(1, 0, 0) = M(1, 0 + 0 + 0 \cdot 0, 1) = M(1, 0, 1).$$

Since $M(1, 1, 1) \neq M(1, 0, 1)$, H is not abelian.

- (2) **Proposition.** Let G_1, G_2 be groups with identities e_1, e_2 respectively, and let $\varphi : G_1 \rightarrow G_2$ be a homomorphism. Then

$$\varphi(e_1) = e_2, \quad \varphi(a^{-1}) = \varphi(a)^{-1} \quad \text{for all } a \in G_1.$$

Proof.

$\varphi(e_1) = e_2$. Since $e_1 e_1 = e_1$ in G_1 , applying φ to both sides and using the homomorphism property gives

$$\varphi(e_1)\varphi(e_1) = \varphi(e_1).$$

Multiplying both sides on the left by $\varphi(e_1)^{-1}$ (which exists since G_2 is a group) cancels the extra $\varphi(e_1)$ on the left, leaving

$$\varphi(e_1) = e_2.$$

$\varphi(a^{-1}) = \varphi(a)^{-1}$. Fix $a \in G_1$. Since $aa^{-1} = e_1$ and $a^{-1}a = e_1$ in G_1 ,

$$\varphi(a)\varphi(a^{-1}) = \varphi(aa^{-1}) = \varphi(e_1) = e_2, \quad \varphi(a^{-1})\varphi(a) = \varphi(a^{-1}a) = \varphi(e_1) = e_2,$$

using $\varphi(e_1) = e_2$ from above. So $\varphi(a^{-1})$ is a two-sided inverse of $\varphi(a)$ in G_2 . Since inverses in a group are unique, $\varphi(a^{-1}) = \varphi(a)^{-1}$.

- (3) **Proposition.** The map $\varphi : H \rightarrow G$, $G = \mathbb{R} \times \mathbb{R} \times S^1$, given by

$$\varphi \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} = (a, c, e^{ib}),$$

is surjective. Moreover, with the multiplication

$$(x_1, y_1, u_1) \cdot (x_2, y_2, u_2) = (x_1 + x_2, y_1 + y_2, e^{ix_1 y_2} u_1 u_2),$$

G is a group, and φ is a group homomorphism from H onto G .

Proof.

Surjectivity. Let $(x, y, u) \in G$ be arbitrary, so $x, y \in \mathbb{R}$ and $u \in S^1$. Since $u \in S^1$, $u = e^{i\theta}$ for some $\theta \in \mathbb{R}$. Let

$$M := \begin{pmatrix} 1 & x & \theta \\ 0 & 1 & y \\ 0 & 0 & 1 \end{pmatrix} \in H.$$

Then, by definition of φ (with $a = x$, $b = \theta$, $c = y$),

$$\varphi(M) = (x, y, e^{i\theta}) = (x, y, u).$$

Since $(x, y, u) \in G$ was arbitrary, φ is surjective.

G is a group.

Closure. $x_1 + x_2, y_1 + y_2 \in \mathbb{R}$. Since $|e^{ix_1 y_2}| = |u_1| = |u_2| = 1$, the product $e^{ix_1 y_2} u_1 u_2$ has modulus 1, so it lies in S^1 . Hence the product lies in G .

Identity. Take $e = (0, 0, 1)$. Then

$$(x, y, u) \cdot (0, 0, 1) = (x + 0, y + 0, e^{ix \cdot 0} u \cdot 1) = (x, y, u), \quad (0, 0, 1) \cdot (x, y, u) = (x, y, u).$$

So $(0, 0, 1)$ is a two-sided identity.

Associativity. Let $A = (x_1, y_1, u_1)$, $B = (x_2, y_2, u_2)$, $C = (x_3, y_3, u_3)$. Computing AB first,

$$(AB)C = (x_1 + x_2 + x_3, y_1 + y_2 + y_3, e^{i(x_1 y_2 + x_1 y_3 + x_2 y_3)} u_1 u_2 u_3),$$

and computing BC first,

$$A(BC) = (x_1 + x_2 + x_3, y_1 + y_2 + y_3, e^{i(x_1 y_2 + x_1 y_3 + x_2 y_3)} u_1 u_2 u_3).$$

These agree, so $(AB)C = A(BC)$.

Inverses. Given $(x, y, u) \in G$, seek (x', y', u') with $(x, y, u) \cdot (x', y', u') = (0, 0, 1)$. This requires $x' = -x$, $y' = -y$, and

$$e^{ixy'} u u' = 1 \implies u' = e^{ixy} u^{-1}.$$

Checking the other order confirms $(-x, -y, e^{ixy} u^{-1}) \cdot (x, y, u) = (0, 0, 1)$ as well, so $(x, y, u)^{-1} = (-x, -y, e^{ixy} u^{-1}) \in G$ is a two-sided inverse.

φ is a homomorphism. For $M = M(a, b, c)$, $M' = M(a', b', c') \in H$, the multiplication law $(*)$ from B7(1) gives $MM' = M(a + a', b + b' + ac', c + c')$, so

$$\varphi(MM') = (a + a', c + c', e^{i(b+b'+ac')}).$$

On the other hand,

$$\varphi(M)\varphi(M') = (a, c, e^{ib}) \cdot (a', c', e^{ib'}) = (a + a', c + c', e^{i(ac'+b+b')}).$$

Since $ac' + b + b' = b + b' + ac'$, $\varphi(MM') = \varphi(M)\varphi(M')$.

Combining the three parts: φ is a surjective homomorphism from H onto the group G .

(4) **Proposition.** The kernel of $\varphi : H \rightarrow G$ from B7(3) is

$$\text{Ker}(\varphi) = \left\{ \begin{pmatrix} 1 & 0 & 2\pi k \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} : k \in \mathbb{Z} \right\},$$

and $\text{Ker}(\varphi)$ is a subgroup of H .

Proof.

Explicit form. $M(a, b, c) \in \text{Ker}(\varphi)$ iff $\varphi(M(a, b, c)) = (a, c, e^{ib})$ equals the identity $(0, 0, 1)$ of G , i.e. $a = 0$, $c = 0$, and $e^{ib} = 1$. Since $e^{ib} = 1$ holds exactly when $b = 2\pi k$ for some $k \in \mathbb{Z}$, this gives the claimed set.

Subgroup: nonempty. $\varphi(I_3) = \varphi(M(0, 0, 0)) = (0, 0, e^{i0}) = (0, 0, 1)$, so $I_3 \in \text{Ker}(\varphi)$.

Subgroup: closed under multiplication. Let $M, M' \in \text{Ker}(\varphi)$, so $\varphi(M) = \varphi(M') = (0, 0, 1)$. Since φ is a homomorphism,

$$\varphi(MM') = \varphi(M)\varphi(M') = (0, 0, 1) \cdot (0, 0, 1) = (0, 0, 1),$$

so $MM' \in \text{Ker}(\varphi)$.

Subgroup: closed under inverses. Let $M \in \text{Ker}(\varphi)$, so $\varphi(M) = (0, 0, 1)$. By B7(2),

$$\varphi(M^{-1}) = \varphi(M)^{-1} = (0, 0, 1)^{-1} = (0, 0, 1),$$

so $M^{-1} \in \text{Ker}(\varphi)$.

Since $\text{Ker}(\varphi)$ is nonempty and closed under multiplication and inverses (with associativity inherited from H), $\text{Ker}(\varphi)$ is a subgroup of H .

B8. For $m\mathbb{Z}$ to be an abelian subgroup, we check that the binary operator $+$ satisfies G1, G2, G3, and Ab.

G1, associativity: Let $a, b, c \in m\mathbb{Z}$. Then, $a = ma_1, b = mb_1, c = mc_1$ for some $a_1, b_1, c_1 \in \mathbb{Z}$. We can then note that

$$\begin{aligned} a + (b + c) &= ma_1 + (mb_1 + mc_1) \\ &= (ma_1 + mb_1) + mc_1 && \text{(associativity of integers, integers closed under multiplication)} \\ &= (a + b) + c \end{aligned}$$

G2, identity: Claim that the identity element e is 0, noting that $e = 0 = m \times 0 \implies e \in m\mathbb{Z}$.

$$\begin{aligned} a + e &= ma_1 + m0 = ma_1 + 0 = ma_1 = a \\ e + a &= m0 + ma_1 = 0 + ma_1 = ma_1 = a \\ &\implies a + e = e + a = a \end{aligned}$$

G3, inverse: Claim the inverse element of $a \in m\mathbb{Z}$ is $-a = m(-a_1)$. We note that $-a \in m\mathbb{Z}$ because $a_1 \in \mathbb{Z}$ so $-a_1 \in \mathbb{Z}$. Let's show that these are inverses:

$$\begin{aligned} a + (-a) &= ma_1 + (m(-a_1)) \\ &= ma_1 - ma_1 && \text{(integer arithmetic)} \\ &= 0 \\ &= e \end{aligned}$$

. Similarly,

$$\begin{aligned} (-a) + a &= m(-a_1) + ma_1 \\ &= -ma_1 + ma_1 \\ &= 0 \\ &= e \end{aligned}$$

Ab, commutativity:

$$\begin{aligned} a + b &= ma_1 + mb_1 \\ &= mb_1 + ma_1 && \text{(commutativity of } \mathbb{Z} \text{)} \\ &= b + a \end{aligned}$$

Also I guess we included that there was closure, because $a + b = ma_1 + mb_1 = m(a_1 + b_1) \in m\mathbb{Z}$ because $a_1 + b_1 \in \mathbb{Z}$.

(1) Let $f : m\mathbb{Z} \rightarrow \mathbb{Z}$ be the invertible homomorphism. We will define f as $f(x) = m^{-1}x$. This also means that $f^{-1}(x) = mx$. We can note that this is a bijection because:

injectivity: If $f(a) = f(b)$, ($a = ma_1, b = mb_1$), then $a/m = b/m \implies ma_1/m = mb_1/m \implies a_1 = b_1 \implies ma_1 = mb_1 \implies a = b$, so f is injective.

surjectivity: Let $y \in \mathbb{Z}$ be arbitrary. We claim that $x = my$ is the input such that $f(x) = y$. First note that $x = my \in m\mathbb{Z}$ since $y \in \mathbb{Z}$. Then,

$f(x) = f(my) = m^{-1}my = y$, so f is surjective.

Homomorphism: Let $a, b \in m\mathbb{Z}$, such that $a = ma_1, b = mb_1$, and $a_1, b_1 \in \mathbb{Z}$.

$$\begin{aligned}
f(a+b) &= f(ma_1 + mb_1) \\
&= f(m(a_1 + b_1)) && \text{(regular integer distributivity)} \\
&= f(mc_1) && \text{(for } c_1 := a_1 + b_1 \in \mathbb{Z}) \\
&= f(c) && \text{(where } c = mc_1 \in m\mathbb{Z}) \\
&= m^{-1}c \\
&= m^{-1}mc_1 \\
&= c_1 \\
&= a_1 + b_1 \\
&= m^{-1}ma_1 + m^{-1}mb_1 \\
&= f(ma_1) + f(mb_1) \\
&= f(a) + f(b)
\end{aligned}$$

(2) **i is a group homomorphism:** Let $a, b \in m\mathbb{Z}$ such that $a = ma_1, b = mb_1$. Then, $i(a+b) = a+b = i(a) + i(b)$.

prove no good p : Consider arbitrary p . We consider input $m \in m\mathbb{Z}$. Then,

$$\begin{aligned}
(p \circ i)(m) &= p(i(m)) \\
&= p(m) \\
&= p(1 + \cdots + 1) && (m \text{ times}) \\
&= p(1) + \cdots + p(1) && \text{(homomorphism property applied } m-1 \text{ times)} \\
&= mp(1) \\
&= m && \text{(because we assume } p \circ i = \text{id)}
\end{aligned}$$

Now we have that $mp(1) = m$, so $p(1) = 1$. But $1 \notin m\mathbb{Z}$ unless $m = 1$, since $1 = m * m^{-1}$, and $m^{-1} \notin \mathbb{Z}$ if $m \neq 1$.

Therefore unless $m = 1$, we have a contradiction. So this p is not valid.